Permutations and Combinations

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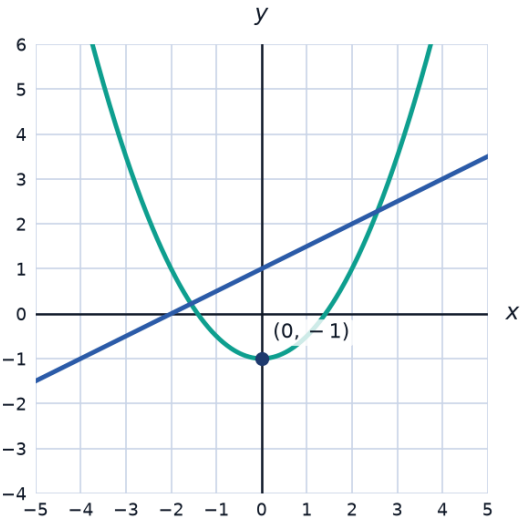
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word COMMITTEE.

2 [Foundational] • Target: Fundamental Counting and Linear Permutations

A group of n students, which includes Ali and Priya, are to be arranged in a straight line for a photograph. The number of different arrangements in which Ali and Priya are not standing next to each other is 3600. Find the value of the integer n.



3 [Foundational] • Target: Permutations with Repetitions and Identical Items

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

4 [Foundational] • Target: Combinations and Committee Selections

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed? Show your complete algebraic work.

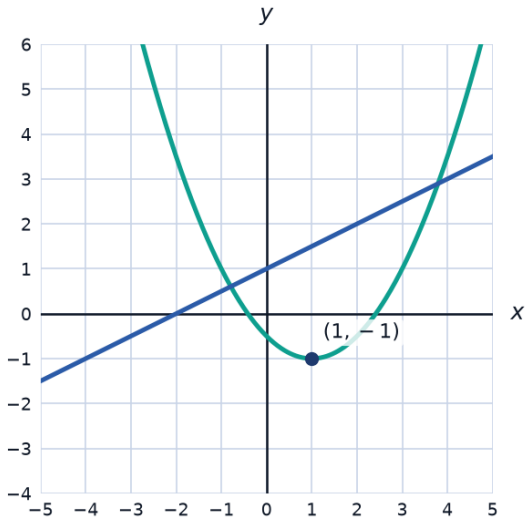
5 [Foundational] • Target: Restricted Arrangements and Separations

Find the number of distinct arrangements of the letters of the word SUCCESSFULLY.

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6 [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word ENGINEERING.



7 [Foundational] • Target: Fundamental Counting and Linear Permutations

Find the number of distinct arrangements of the letters of the word ARRANGEMENT.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Permutations with Repetitions and Identical Items

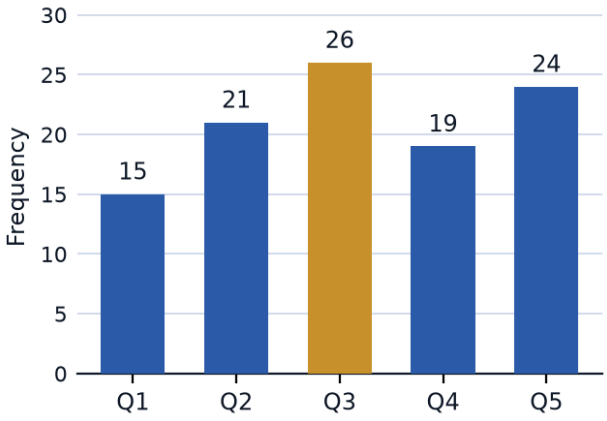
Find the number of distinct arrangements of the letters of the word MATHEMATICS.

9 [Intermediate] • Target: Combinations and Committee Selections

Find the number of distinct arrangements of the letters of the word ACCOUNTANT.

10 [Intermediate] • Target: Restricted Arrangements and Separations

In how many distinct ways can 12 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.



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11 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word MISSISSIPPI.

12 [Intermediate] • Target: Fundamental Counting and Linear Permutations

Find the number of distinct arrangements of the letters of the word STATISTICS.

13 [Intermediate] • Target: Permutations with Repetitions and Identical Items

In how many distinct ways can 11 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

14 [Intermediate] • Target: Combinations and Committee Selections

In how many distinct ways can 10 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

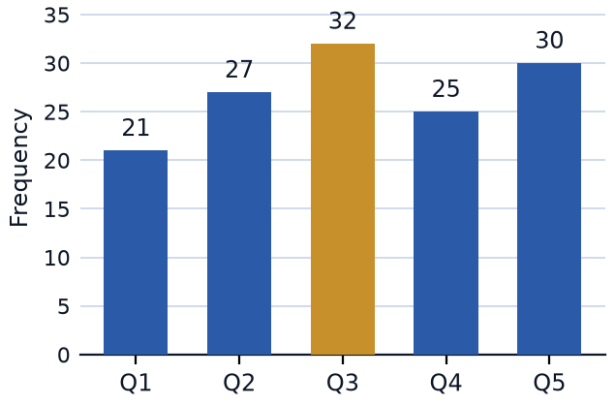
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the number of distinct arrangements of the letters of the word ARRANGE.

16 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

In how many distinct ways can 9 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.



17 [Challenging] • Target: Fundamental Counting and Linear Permutations • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the number of distinct arrangements of the letters of the word BALLOON.

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18 [Challenging] • Target: Permutations with Repetitions and Identical Items • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the number of distinct arrangements of the letters of the word ADDRESS.

19 [Challenging] • Target: Combinations and Committee Selections • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In how many distinct ways can 8 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

20 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations $ {}^{5}P_{2} $.

a. 25

b. 21

c. 10

d. 20

22 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations $ {}^{6}P_{2} $.

a. 12

b. 15

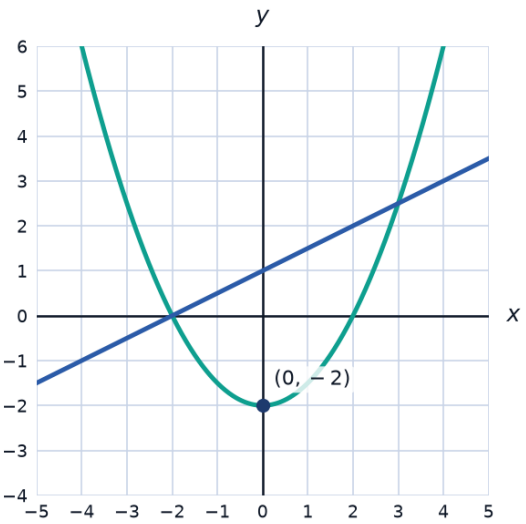
c. 36

d. 30

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23 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{7}P_{2} \).



a. 21

b. 42

c. 49

d. 14

24 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{8}P_{2} \).

a. 64

b. 56

c. 16

d. 28

25 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{9}P_{2} \).

a. 72

b. 81

c. 18

d. 36

26 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{10}P_{2} \).

a. 90

b. 100

c. 20

d. 45

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27 [Foundational] • Target: Permutations & Combinations Practice Exercises

A committee of 4 people is to be chosen from \(5 \) men and \(4 \) women. How many committees consist of exactly 2 men and 2 women?

a. 60

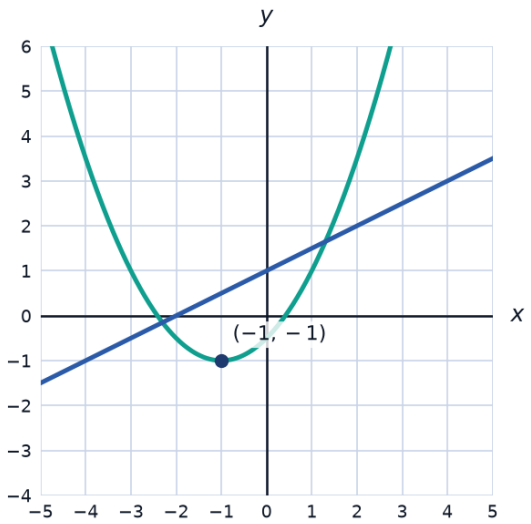
b. 120

c. 55

d. 36

28 [Foundational] • Target: Permutations & Combinations Practice Exercises

A committee of 4 people is to be chosen from \(6 \) men and \(5 \) women. How many committees consist of exactly 2 men and 2 women?



a. 44

b. 145

c. 300

d. 150

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Fundamental Counting and Linear Permutations

In how many ways can 5 different books be arranged on a shelf?

a. 120

b. 24

c. 60

d. 720

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30 [Intermediate] • Target: Fundamental Counting and Linear Permutations

Evaluate the number of permutations of 7 distinct items taken 3 at a time, ${}^{7}P_{3}$.

a. 210

b. 35

c. 42

d. 840

31 [Intermediate] • Target: Fundamental Counting and Linear Permutations

How many 4-digit even numbers can be formed using digits $\{1, 2, 3, 5, 7, 8\}$ without repetition?

a. 120

b. 60

c. 240

d. 180

32 [Intermediate] • Target: Permutations with Repetitions and Identical Items

Find the number of distinct permutations of all the letters in the word PARALLEL.

a. 3360

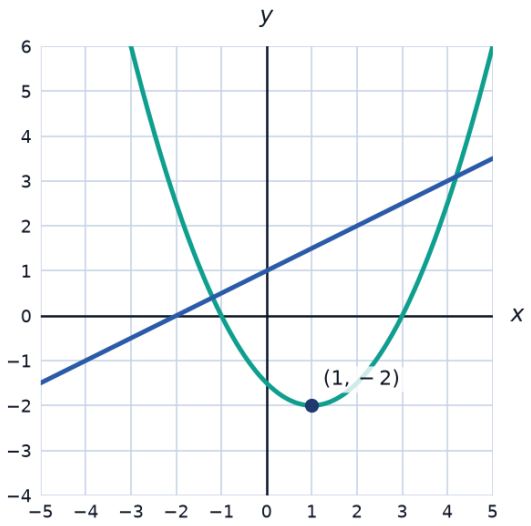
b. 6720

c. 1680

d. 40320

33 [Intermediate] • Target: Permutations with Repetitions and Identical Items

How many 6-letter arrangements of PEPPER begin with the letter P?



a. 30

b. 60

c. 20

d. 120

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34 [Intermediate] • Target: Permutations with Repetitions and Identical Items

How many 5-digit codes can be formed from digits $\{3, 3, 3, 7, 7\}$?

a. 10

b. 20

c. 5

d. 120

35 [Intermediate] • Target: Combinations and Committee Selections

A committee of 4 is to be chosen from 9 people. How many distinct committees can be formed?

a. 126

b. 3024

c. 84

d. 252

36 [Intermediate] • Target: Combinations and Committee Selections

A team of 5 is selected from 6 boys and 4 girls. How many teams contain exactly 3 boys and 2 girls?

a. 120

b. 60

c. 210

d. 100

37 [Intermediate] • Target: Combinations and Committee Selections

From 10 points on a plane where no three are collinear, how many distinct triangles can be drawn?

a. 120

b. 720

c. 90

d. 45

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In how many ways can 4 boys and 3 girls sit in a row if all 3 girls must sit together?

a. 720

b. 5040

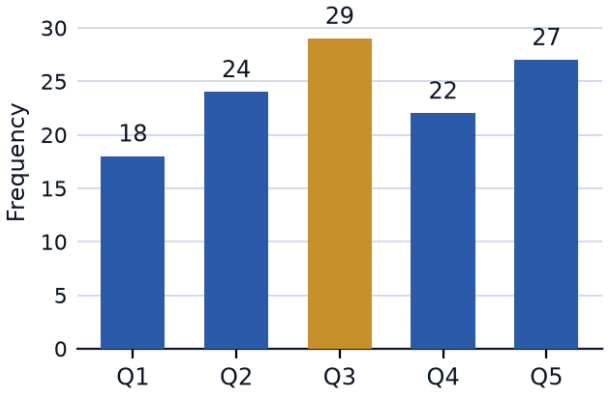
c. 144

d. 288

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39 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In how many ways can 5 people sit around a circular dinner table?



a. 24

b. 120

c. 60

d. 12

40 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A committee of 4 is selected from 5 men and 5 women. How many committees contain at least one woman?

a. 205

b. 210

c. 195

d. 150

41 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

In how many distinct ways can 7 different people be seated around a circular table (rotations considered identical)?

a. 720

b. 766

c. 647

d. 611

42 [Challenging] • Target: Fundamental Counting and Linear Permutations • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the number of distinct arrangements of the letters of the word GIGGLE.

a. 214

b. 141

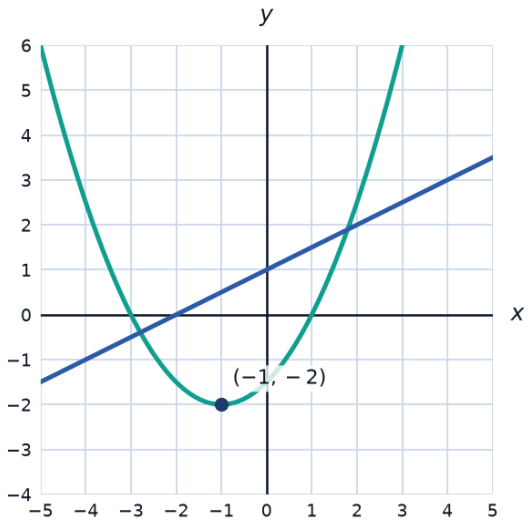
c. 180

d. 207

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43 [Challenging] • Target: Permutations with Repetitions and Identical Items • Cross-Lesson: 8.01 Integration and Kinematics

Find the number of distinct arrangements of the letters of the word LEVEL.



a. 12

b. 29

c. 30

d. 76

44 [Challenging] • Target: Combinations and Committee Selections • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the number of distinct arrangements of the letters of the word PEPPER.

a. 33

b. 17

c. 34

d. 60

45 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the number of distinct arrangements of the letters of the word BANANA.

a. 52

b. 71

c. 60

d. 62

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